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(54) NOVEL HIGH-SPEED VIA IMPEDANCE MATCHING METHODOLOGY USING LOW DIELECTRIC CONSTANT BACKDRILL FILLING MATERIAL

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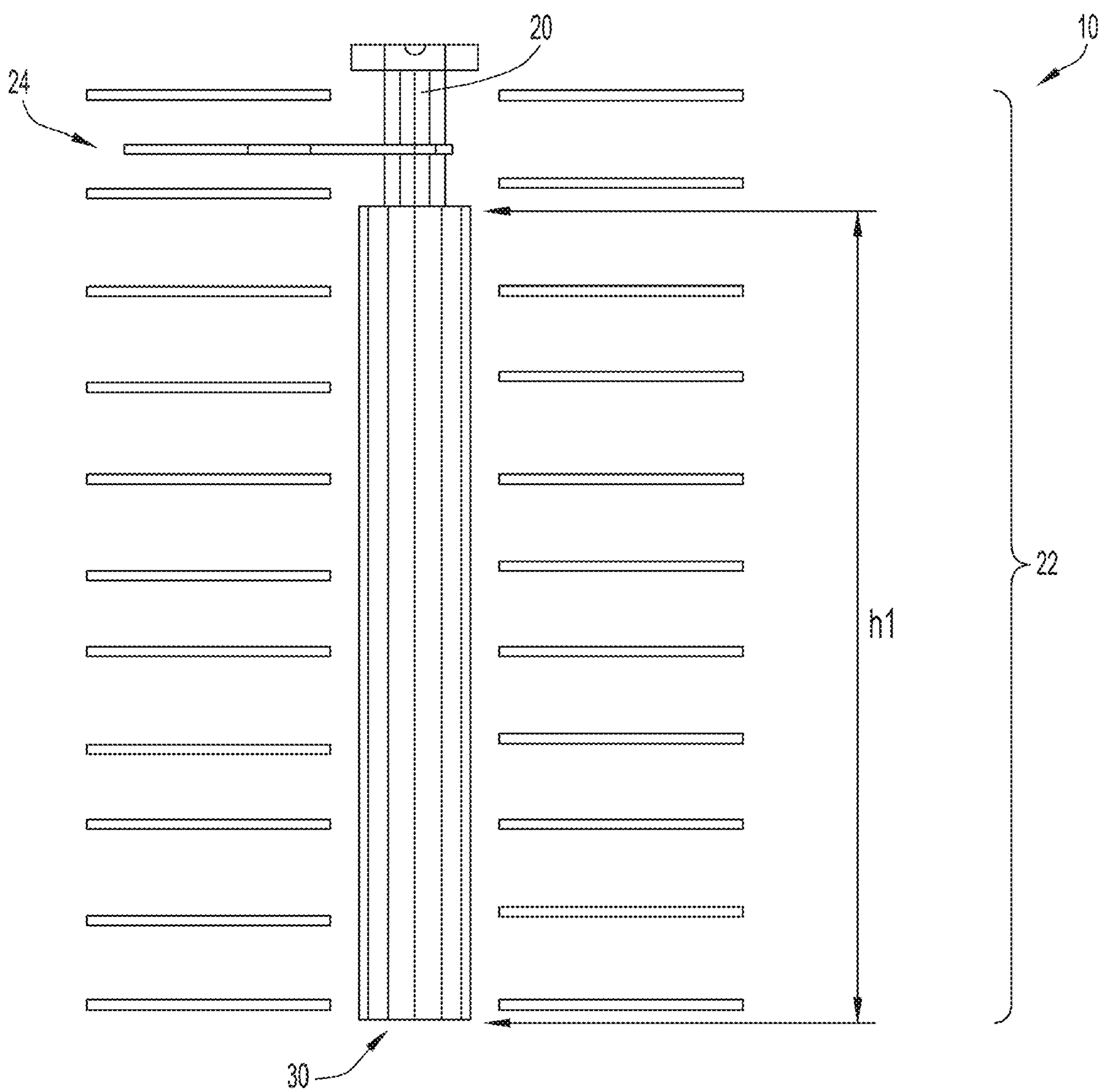
(52) **U.S. Cl.**
CPC ***H05K 1/0251*** (2013.01); ***H05K 1/024*** (2013.01); *H05K 2201/09627* (2013.01)

(57) **ABSTRACT**

A process for improving or reducing impedance mismatches between VIAs and traces for signal frequencies at 56 GHZ and greater by using a lower dk material for backdrill filling. By reducing such impedance mismatches, which in other words results in reducing such mismatches, the performance result is a reduction in signal reflections. In some cases, the signal reflection reduction is significant.



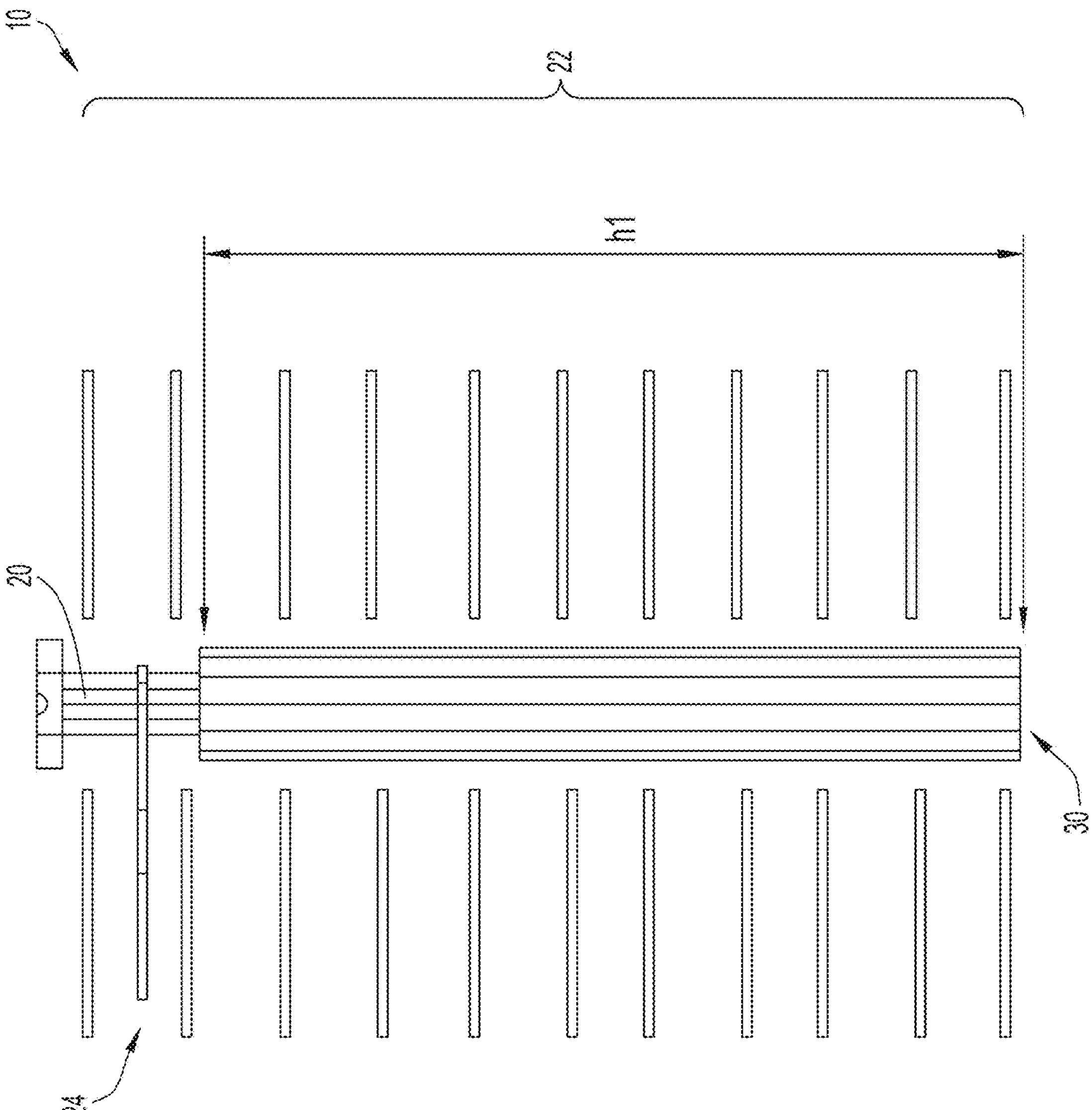


FIG.1

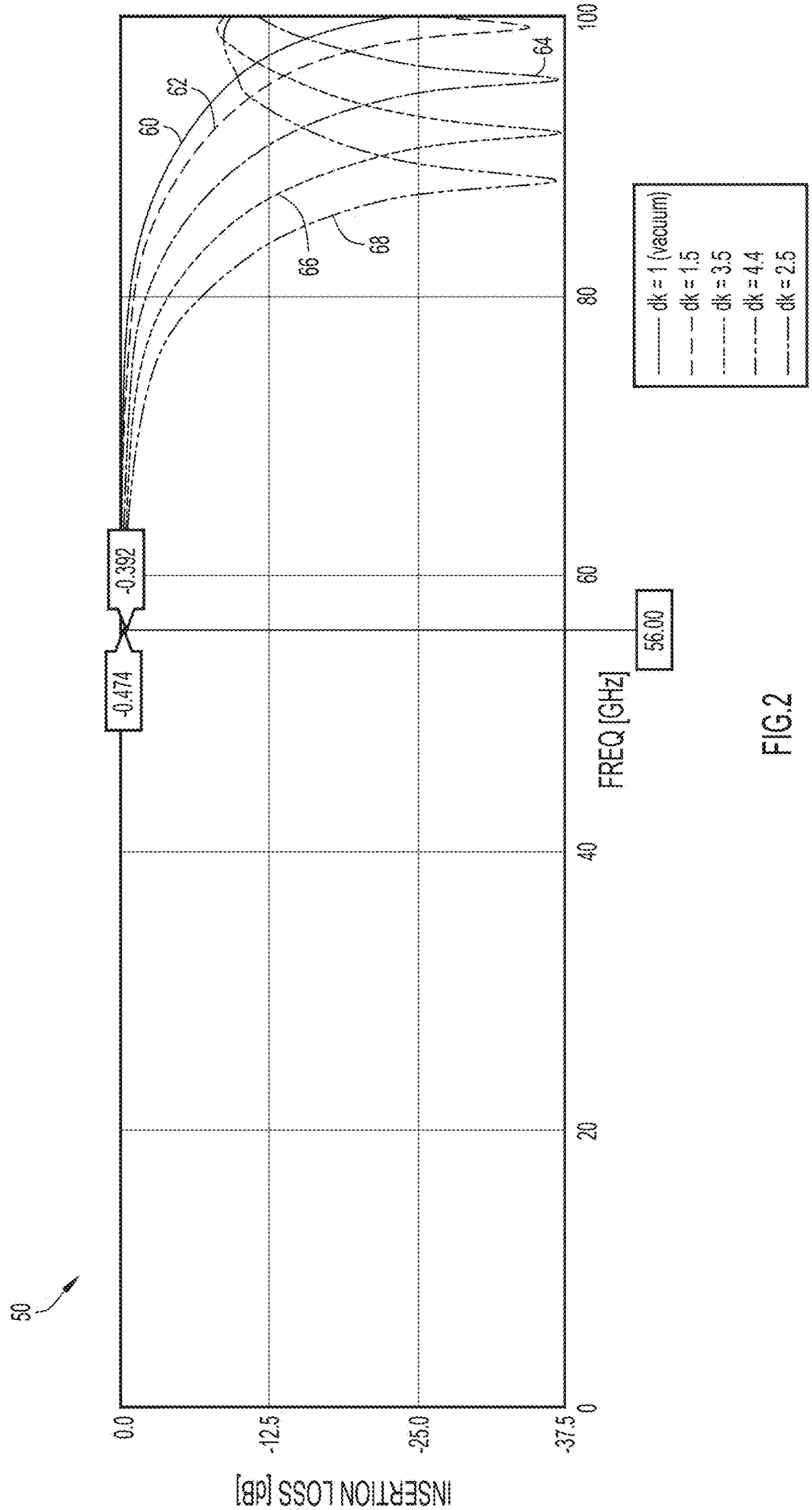

50
INSERTION LOSS [dB]
0.0
-12.5
-25.0
-37.5
FREQ [GHz]
0
20
40
60
80
100
-0.474
-0.392
56.00
60
62
64
66
68
dk = 1 (vacuum)
dk = 1.5
dk = 3.5
dk = 4.4
dk = 2.5

FIG.2

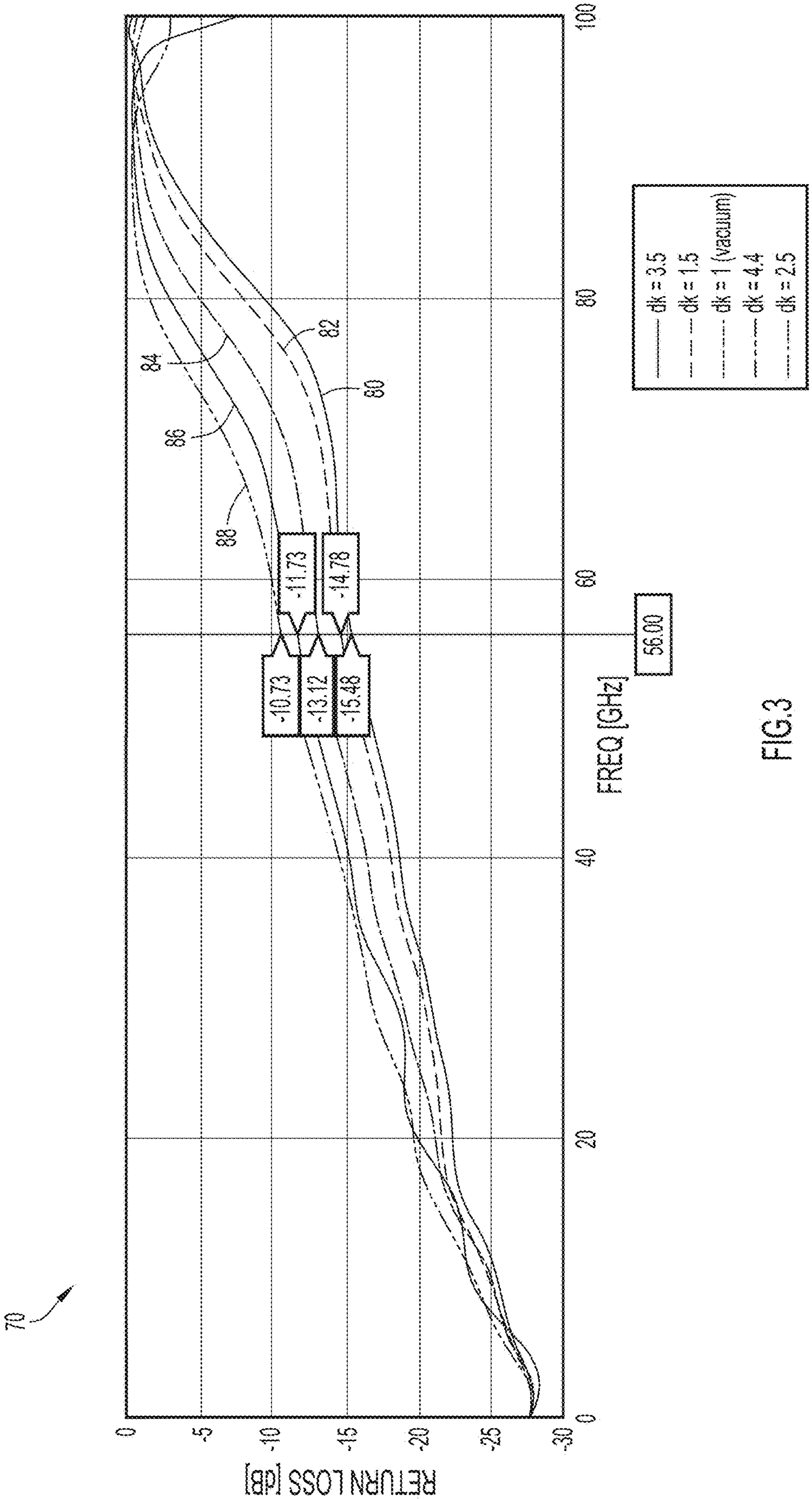

70
RETURN LOSS [dB]
0
-5
-10
-15
-20
-25
-30
FREQ [GHz]
0
20
40
60
80
100
56.00
-10.73
-13.12
-15.48
-11.73
-14.78
80
82
84
86
88
dk = 3.5
dk = 1.5
dk = 1 (vacuum)
dk = 4.4
dk = 2.5

FIG.3

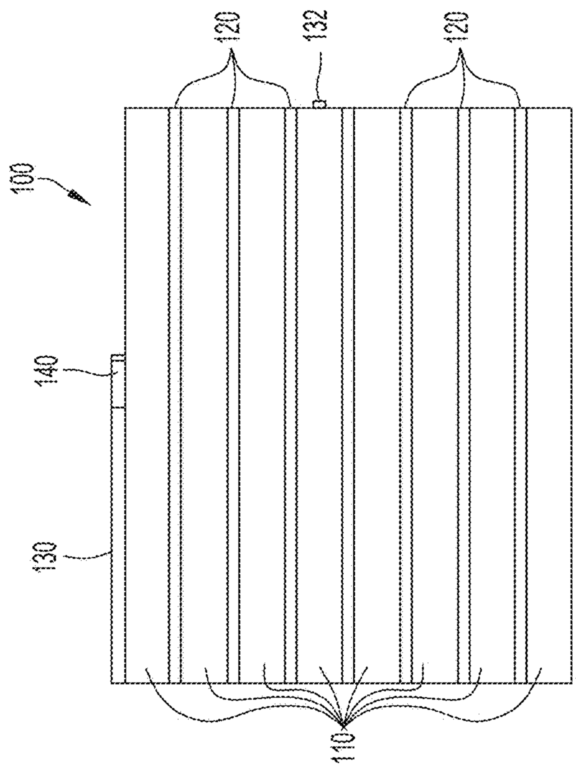


FIG.4B

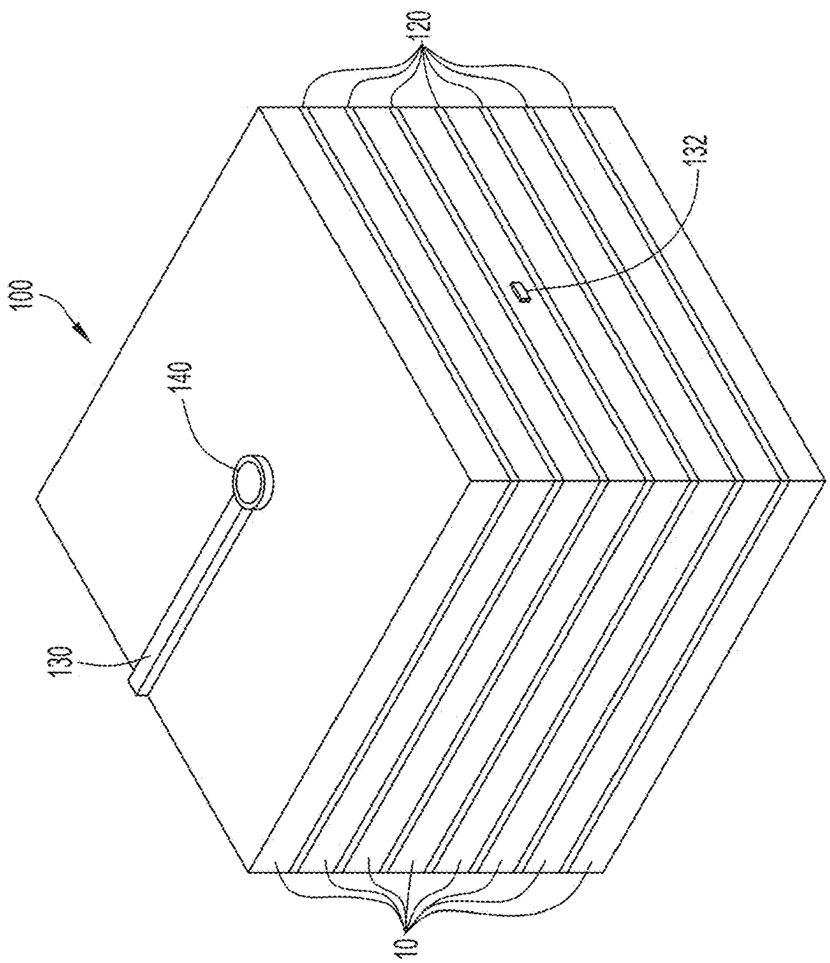


FIG.4A

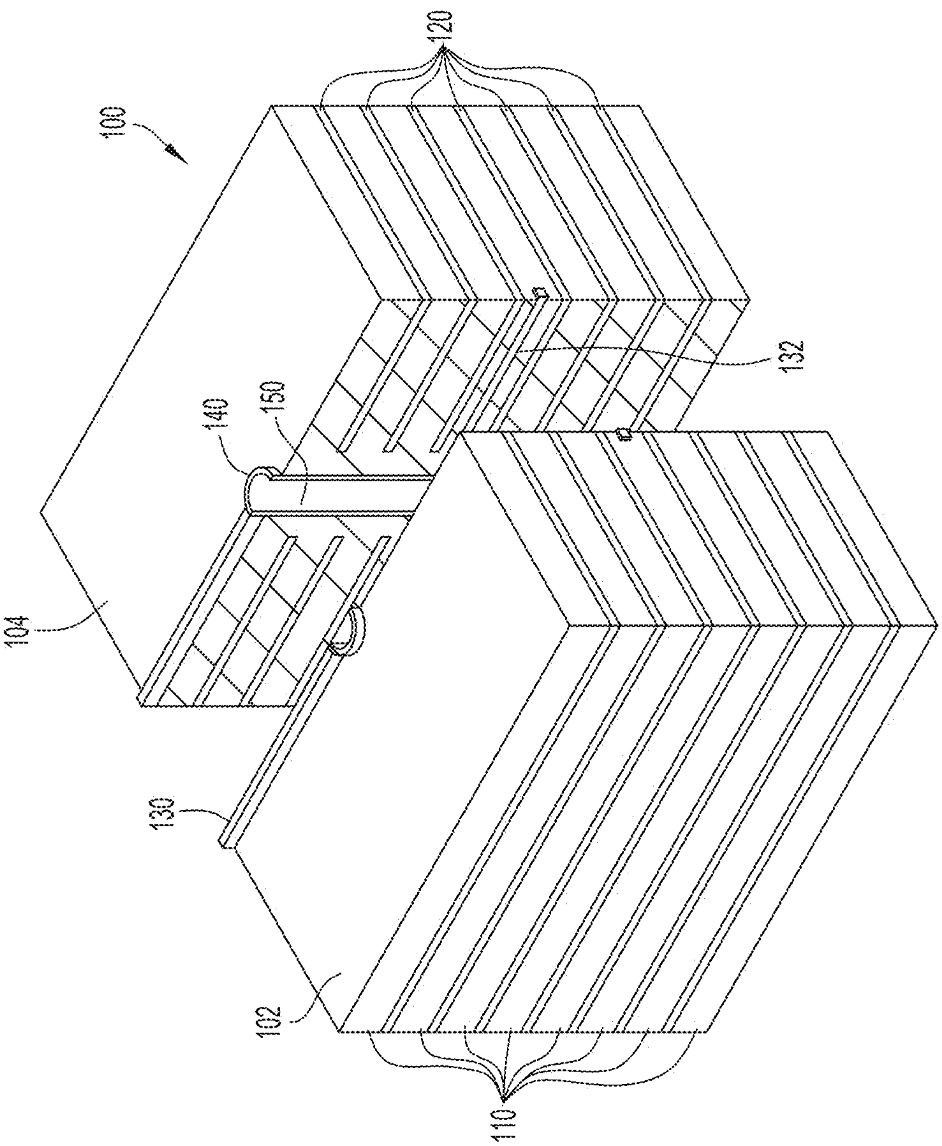


FIG.5

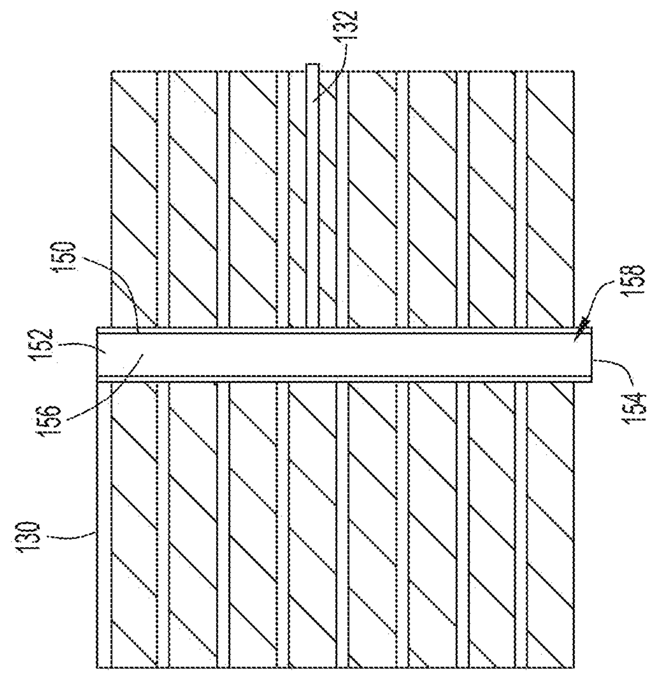

132
150
152
156
130
158
154

FIG.6B

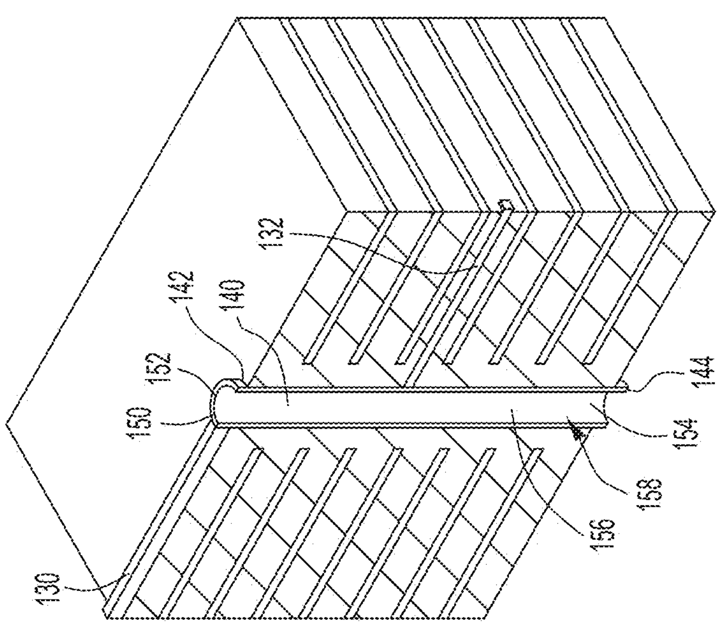

132
142
140
152
150
144
154
158
156
130

FIG.6A

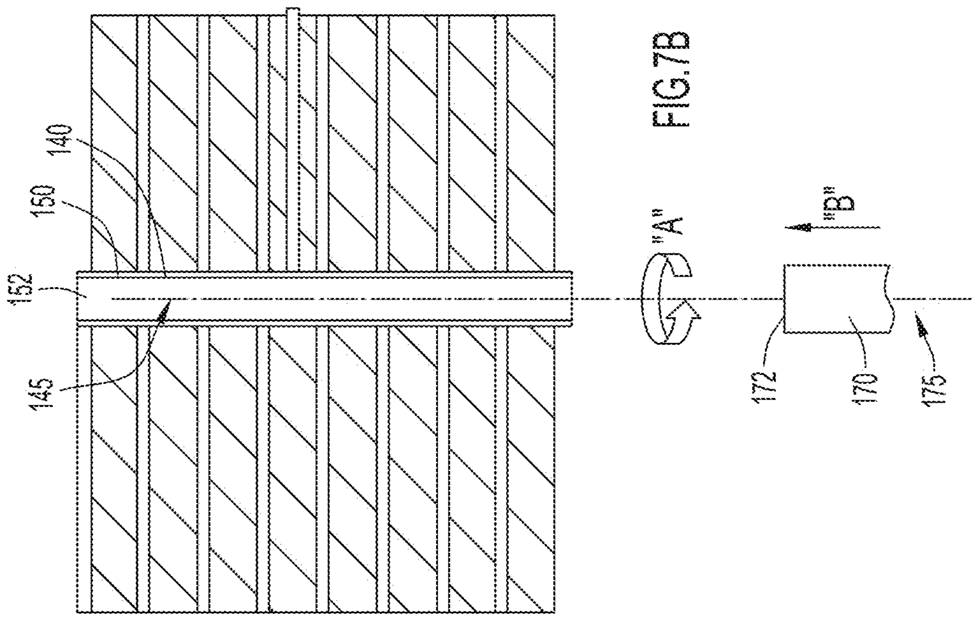


FIG.7B

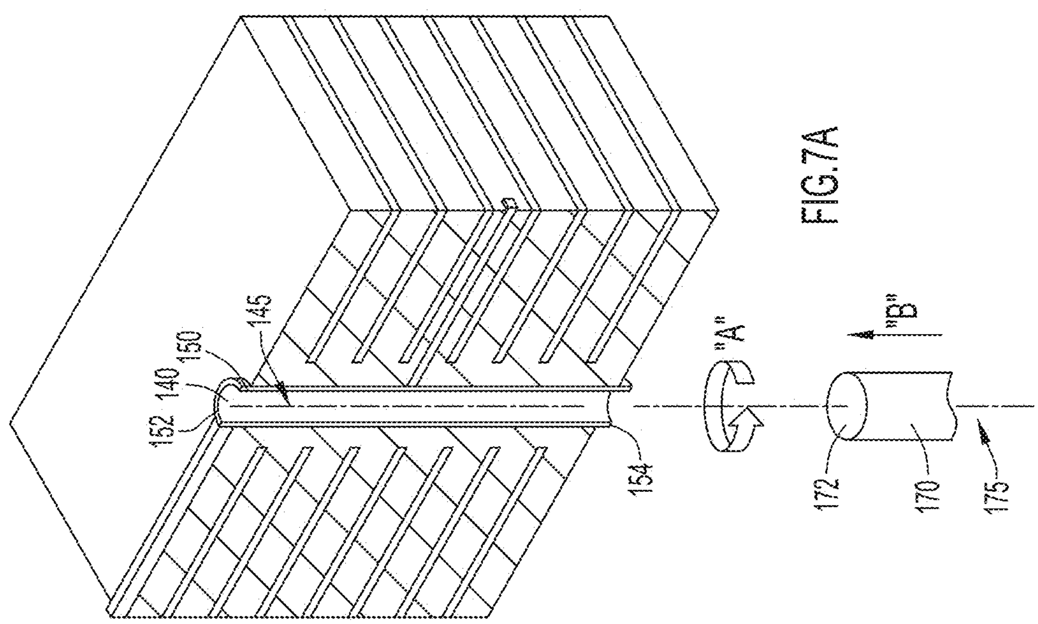


FIG.7A

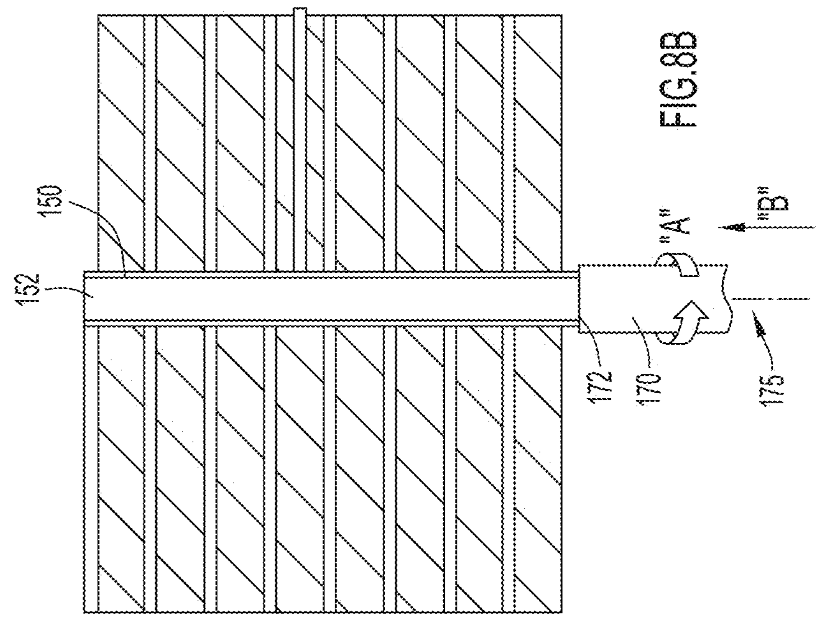


FIG.8B

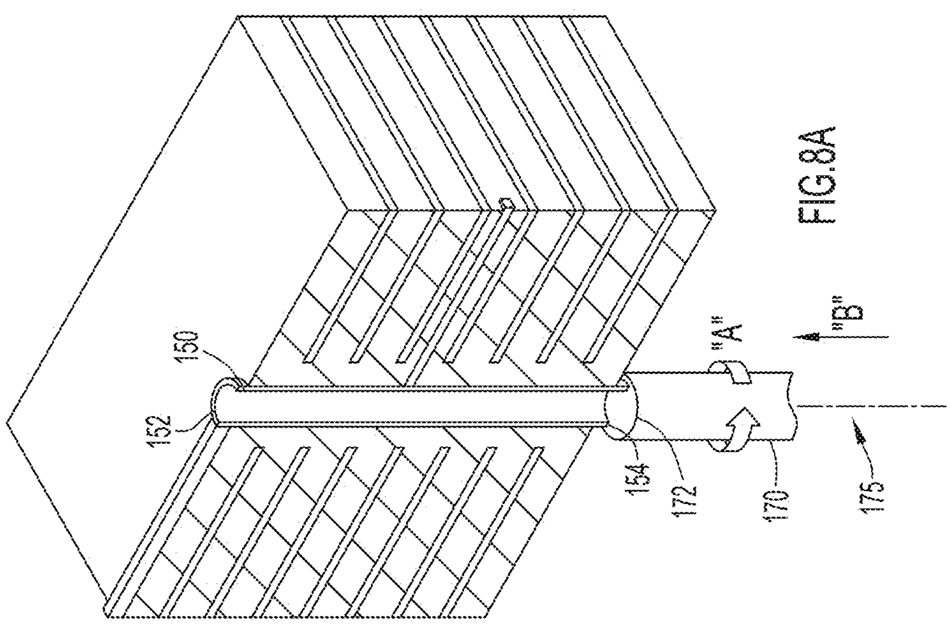


FIG.8A

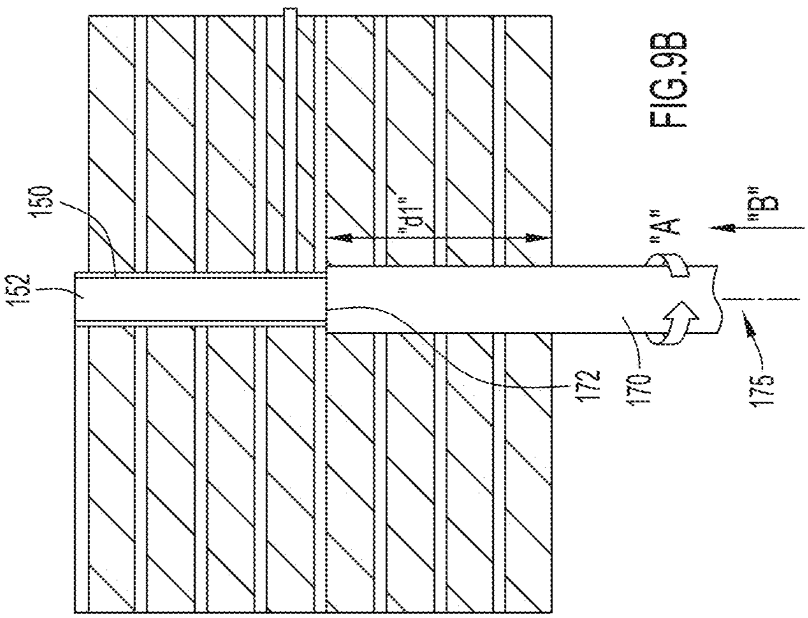


FIG.9B

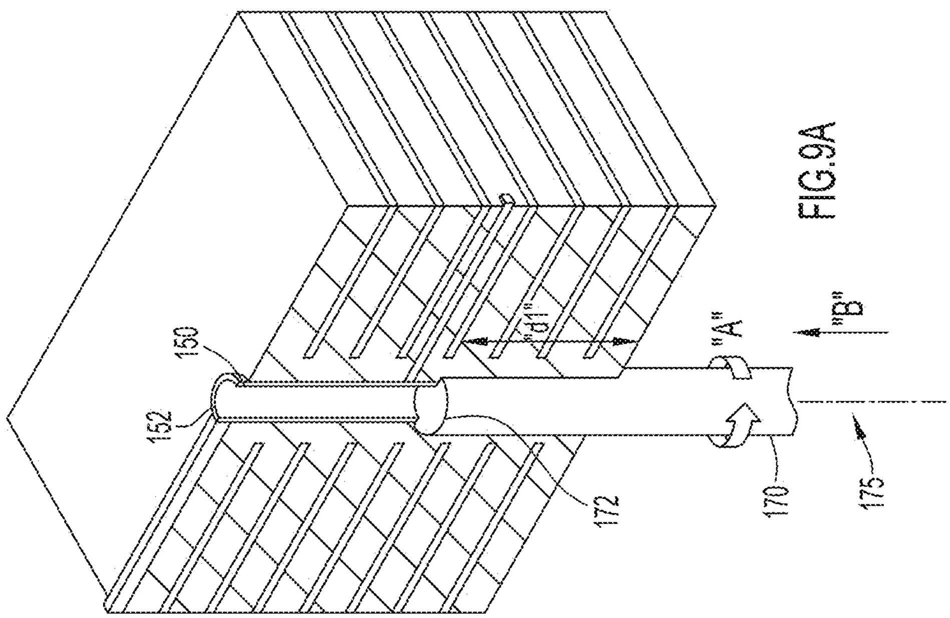


FIG.9A

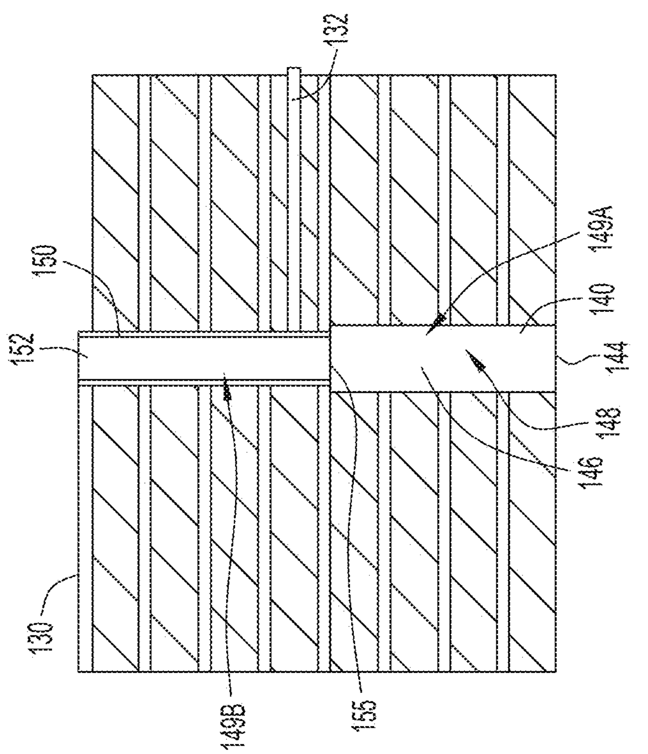


FIG.10B

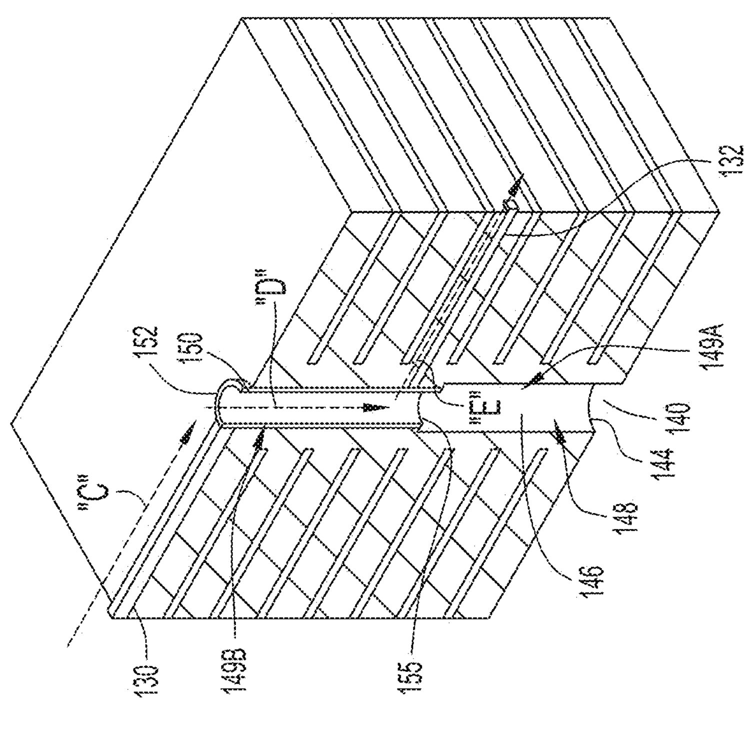


FIG.10A

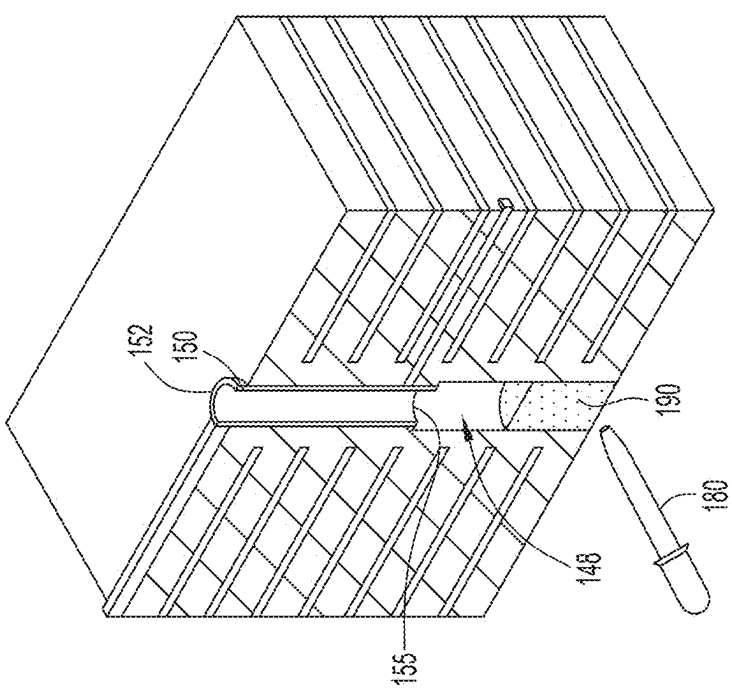


FIG.11A

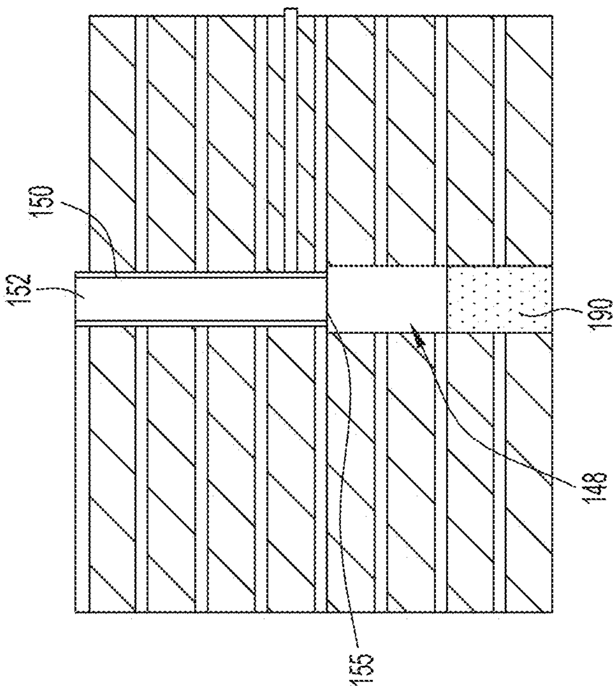


FIG.11B

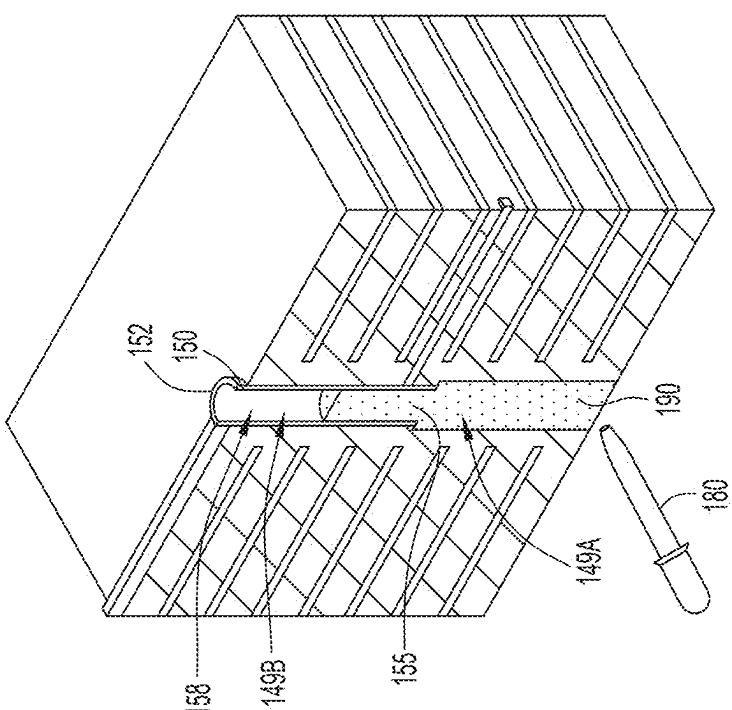


FIG.12A

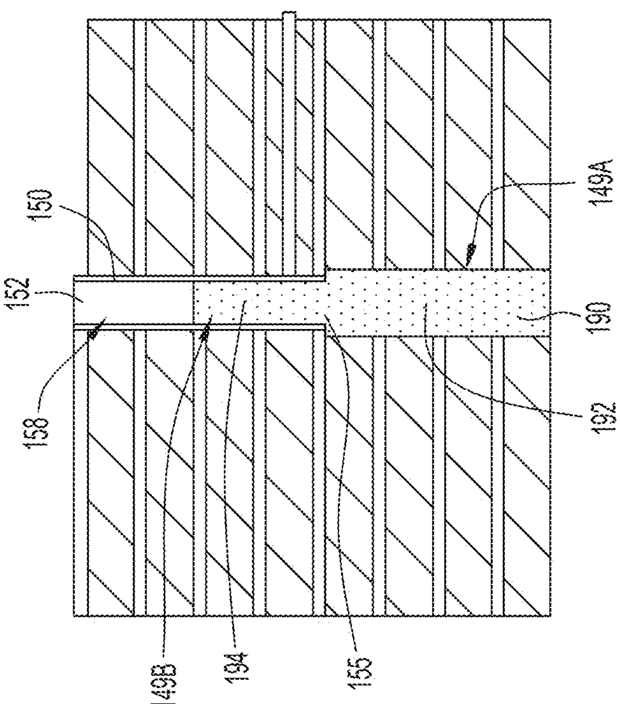


FIG.12B

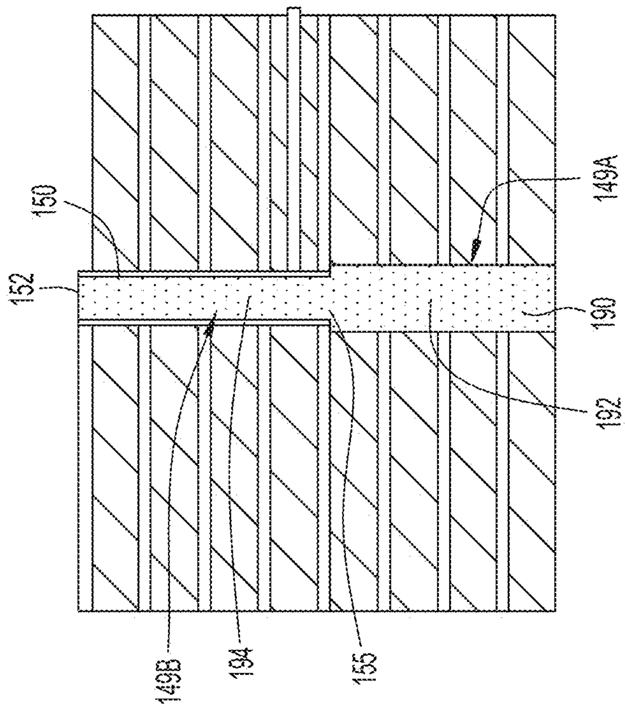


FIG.13B

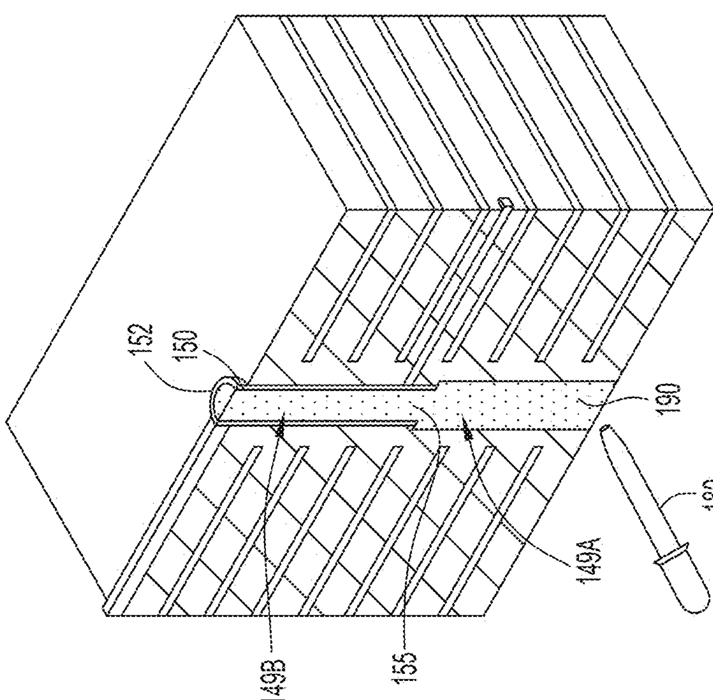


FIG.13A

NOVEL HIGH-SPEED VIA IMPEDANCE MATCHING METHODOLOGY USING LOW DIELECTRIC CONSTANT BACKDRILL FILLING MATERIAL

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a non-provisional application of and claims the benefit of and priority to U.S. Provisional Patent Application No. 63/557,705 filed Feb. 26, 2024, entitled "Novel High-Speed Via Impedance Matching Methodology Using Low Dielectric Constant Backdrill Filling Material", with Attorney Docket No. 0370.7939P/1044405, the entire disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to circuit boards and, more particularly, to circuit boards that have improved impedance matching between traces and VIAs.

BACKGROUND

[0003] The rapid growth of artificial intelligence technologies are driving data centers with higher bandwidth requirements with increasingly higher data rates. With increasing data rates fundamental frequencies increase (for 224 Gbps PAM4 signaling at approximately 56 GHz) and margins are becoming smaller and smaller. One of the performances limiting factors are reflections caused by mismatching of impedances of PCB transition VIAs and traces (connecting to VIAs). These mismatches are causing: 1) high return loss resulting in reflections; and 2) resonances appearing in insertion loss at higher frequencies causing additional reflections.

[0004] Conventional solutions and methodologies do not address large impedance mismatches between VIAs and traces that are caused by high dielectric backdrill material. Thus, there is a need to improve the transition of a VIA to a trace to reduce reflections and avoid a system failure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. **1** is a schematic side view of an embodiment of a printed circuit board (PCB) showing a high-speed VIA according to the techniques disclosed herein.

[0006] FIG. **2** is a graph illustrating the insertion loss of a VIA structure with different backdrill materials having different dielectric constants.

[0007] FIG. **3** is a graph illustrating the return loss of a VIA structure with different backdrill materials having different dielectric constants.

[0008] FIG. **4**A is a top perspective view of an embodiment of a PCB according to the techniques disclosed herein.

[0009] FIG. **4**B is a side view of the PCB illustrated in FIG. **4**A.

[0010] FIG. **5** is a top perspective view of the PCB illustrated in FIG. **4**A having been separated into two pieces to show inner components of the PCB.

[0011] FIG. **6**A is a top perspective view of the PCB illustrated in FIG. **4**A showing an internal view of the PCB.

[0012] FIG. **6**B is a side view of the PCB illustrated in FIG. **6**A.

[0013] FIG. **7**A is a top perspective view of the PCB illustrated in FIG. **6**A showing an embodiment of a drill bit that can be used to backdrill the VIA.

[0014] FIG. **7**B is a side view of the PCB and drill bit illustrated in FIG. **7**A.

[0015] FIG. **8**A is a top perspective view of the PCB illustrated in FIG. **7**A showing the drill bit engaged with the VIA.

[0016] FIG. **8**B is a side view of the PCB and drill bit illustrated in FIG. **8**A.

[0017] FIG. **9**A is a top perspective view of the PCB illustrated in FIG. **8**A showing the drill bit in a position in which the drill bit has been advanced into the VIA and has removed plating from the VIA.

[0018] FIG. **9**B is a side view of the PCB and drill bit illustrated in FIG. **9**A.

[0019] FIG. **10**A is a top perspective view of the PCB illustrated in FIG. **9**A showing the backdrilled VIA.

[0020] FIG. **10**B is a side view of the PCB illustrated in FIG. **10**A.

[0021] FIG. **11**A is a top perspective view of the PCB illustrated in FIG. **10**A showing a supplying device providing a filling material into the backdrilled portion of the VIA.

[0022] FIG. **11**B is a side view of the PCB illustrated in FIG. **11**A.

[0023] FIG. **12**A is a top perspective view of the PCB illustrated in FIG. **11**A showing the filling material in both the backdrilled portion of the VIA and the non-backdrilled portion of the VIA.

[0024] FIG. **12**B is a side view of the PCB illustrated in FIG. **12**A.

[0025] FIG. **13**A is a top perspective view of the PCB illustrated in FIG. **12**A showing the filling material completely filling the VIA.

[0026] FIG. **13**B is a side view of the PCB illustrated in FIG. **13**A.

DETAILED DESCRIPTION

Overview

[0027] The problems with related art PCBs are addressed by the techniques disclosed herein. One of the main contributors to impedance mismatches between VIAs and traces is when a high dielectric constant ("dk") backdrill material is used in a VIA.

Example Embodiments

[0028] In PCB manufacturing, high-speed VIAs are backdrilled and the VIA hole is filled with a high dk material. Referring to FIG. **1**, a cross-sectional schematic view of a high-speed VIA in a PCB is illustrated. In this implementation, the PCB **10** includes a VIA **20**, which can be a high-speed VIA, and several ground layers **22** with core or prepreg layers therebetween. The PCB **10** includes a high-speed trace **24** that is electrically connected to the VIA **20**. The VIA **20** has been backdrilled to a backdrill height "h**1**". In one implementation, the backdrill height "h**1**" is 104 mil. After the backdrilling process, the VIA **20** is filled with a filling material **30**, which in one implementation can be a resinous material.

[0029] Turning to FIG. **2**, a graph illustrating the insertion loss of a VIA structure with backdrill materials having different dielectric constants is shown. The graph **50** shows the measured relationship between insertion loss in dB and signal frequency in GHz. In particular, the graph **50** shows how insertion loss changes when materials with different dk properties are used for backdrill filling. A lower dk material results in improved insertion loss at higher frequencies.

[0030] Graph **50** illustrates five different lines, each of which is a measurement of a different dk material in a VIA that has a diameter of 27.9 mil. In particular, line **60** is for a vacuum (dk=1) in the VIA with no backdrill filling material present. At a frequency of 56 GHz, the measured insertion loss for dk=1 is −0.35 dB. Line **62** is for a material having a dk of 1.5. At a frequency of 56 GHz, the measured insertion loss for the dk of 1.5 material is −0.39 dB. Line **64** is for a material having a dk of 2.5. At a frequency of 56 GHz, the measured insertion loss for the dk of 2.5 material is −0.47 dB. Line **66** is for a material having a dk of 3.5. At a frequency of 56 GHZ, the measured insertion loss for the dk of 3.5 material is −0.57 dB. Line **68** is for a material having a dk of 4.4. At a frequency of 56 GHz, the measured insertion loss for the dk of 4.4 material is −0.67 dB. The foregoing measured insertion loss data is shown in Table 1 below.

TABLE 1

	dk = 4.4	dk = 3.5	dk = 2.5	dk = 1.5	dk = 1 (vacuum)
IL @ 56 GHz (in dB)	−0.67	−0.57	−0.47	−0.39	−0.35

[0031] The details of some of the measurements shown in FIG. **2** are set forth in the following table. As shown in Table 2, as the dk characteristic of the backdrill material or filling material decreases, the resonance frequency increases. While the material having a dk of 4.4 has a measured resonance frequency of 88.3, which reflects a resonance loss, the material having a dk of 1.5 has a measured resonance frequency of 99 GHz, which is equivalent to a vacuum (dk=1) in the VIA, which is when no filling material is present in the VIA.

TABLE 2

	dk = 4.4	dk = 3.5	dk = 2.5	dk = 1.5	dk = 1 (vacuum)
Resonance Frequency (GHz)	88.3	91.7	95.5	99	99

[0032] Turning to FIG. **3**, a graph illustrating the return loss of a VIA structure with backdrill materials having different dielectric constants is shown. The graph **70** shows the measured relationship between return loss in dB and signal frequency in GHz. In particular, the graph **70** shows how return loss changes when materials with different dk properties are used for backdrill filling. A lower dk material results in improved return loss at higher frequencies.

[0033] Graph **70** illustrates five different lines, each of which is a measurement of a different dk material in a VIA that has a diameter of 27.9 mil. In particular, line **80** is for a vacuum (dk=1) in the VIA with no backdrill filling material present. At a frequency of 56 GHz, the measured return loss for dk=1 is −15.48 dB. Line **82** is for a material having a dk of 1.5. At a frequency of 56 GHz, the measured return loss for the dk of 1.5 material is −14.78 dB. Line **84** is for a material having a dk of 2.5. At a frequency of 56 GHZ, the measured return loss for the dk of 2.5 material is −13.12 dB. Line **86** is for a material having a dk of 3.5. At a frequency of 56 GHz, the measured return loss for the dk of 3.5 material is −11.73 dB. Line **88** is for a material having a dk of 4.4. At a frequency of 56 GHz, the measured return loss for the dk of 4.4 material is −10.73. The foregoing measured return loss data is shown in Table 3 below.

TABLE 3

	dk = 4.4	dk = 3.5	dk = 2.5	dk = 1.5	dk = 1 (vacuum)
RL @ 56 GHz (in dB)	−10.73	−11.73	−13.12	−14.78	−15.48

[0034] As shown in FIG. **3**, the return loss is much better when a lower dk material is used as the filling material in the VIA. For example, at approximately 56 GHZ, the return loss is approximately 4 dB to 5 dB better if a lower dk material, such as a material having a dk of 1.5, is used as compared to a higher dk material, such as a material having a dk of 4.4.

[0035] The techniques described herein relate to improving or reducing impedance mismatches between VIAs and traces for signal frequencies at 56 GHZ and greater by using a lower dk material for the backdrill filling material. By reducing such impedance mismatches, the performance result is a reduction in signal reflections. In some cases, the signal reflection reduction is significant.

[0036] Turning to FIGS. **4**A and **4**B, a top perspective view and a side view, respectively, of an embodiment of a PCB according to the techniques disclosed herein are illustrated. In this embodiment, PCB **100** includes several core or prepreg layers **110**, each of which is spaced apart from an adjacent layer **110** by one of several ground or power planes **120**. The layers and planes can be referred to alternatively as planes or layers. PCB **100** includes high-speed traces **130** and **132** that are located at different vertical levels in the PCB **100**. In FIGS. **4**A and **4**B, only an end of high-speed trace **132** is visible.

[0037] In this embodiment, PCB **100** also includes a VIA **140** that extends through layer **110** and plane **120**. The inner part of the VIA **140** includes a non-conductive material, such as air, and an internal plating layer, which is a conductive plating, that extends along the inner surface of the VIA **140**. The conductive plating of the VIA **140** connects each layer of the PCB **100**. In one implementation, the conductive plating is provided by electroplating. In another implementation, the conductive plating is provided by placing metallic cylinders, such as copper cylinders, into the drilled VIA hole.

[0038] Referring to FIG. **5**, a top perspective view of the PCB **100** is illustrated. In FIG. **5**, the PCB **100** has been separated into two pieces **102** and **104**, which are spaced apart from each other to show the inner components of the PCB **100**.

[0039] Turning to FIGS. **6**A and **6**B, a top perspective view and a side view, respectively, of the PCB **100** are illustrated. As shown, the high-speed traces **130** and **132** are visible in FIGS. **6**A and **6**B. In this embodiment, each of the high-speed traces **130** and **132** is connected to the plating of the VIA **140**.

[0040] The VIA **140** has an upper end **142** and a lower end **144** opposite to the upper end **142**. At this stage of the manufacturing process, the VIA **140** has a plating **150** along the inner surface of the VIA **140**. The plating **150** has an upper end **152** and a lower end **154** opposite to the upper end **152**. The plating **150** also has an inner surface **156** extending therealong that defines a channel **158** that extends from the upper end **152** to the lower end **154**. At this point, the channel **158** is filled with air.

[0041] Referring to FIGS. **7**A and **7**B, a top perspective view and a side view, respectively, of the PCB **100** are illustrated. FIGS. **7**A and **7**B illustrate the next manufacturing step of preparing to backdrill a portion of the VIA **140**. The VIA **140** has a longitudinal axis **145** that extends along the VIA **140** from its upper end **142** to its lower end **144**.

[0042] In the step shown in FIGS. **7**A and **7**B, a drilling device **170**, such as a drill bit, is oriented so that a longitudinal axis **175** of the drilling device **170** is aligned with the longitudinal axis **145** of the VIA **140**. The drilling device **170** has a distal end **172** that is positioned so that it approaches the lower end **154** of the plating **150**. The drilling device **170** is rotatable along arrow "A" about longitudinal axis **175**, and can be moved along the direction of arrow "B" toward the VIA **140** by a tool or machine (not shown).

[0043] Turning to FIGS. **8**A and **8**B, a top perspective view and a side view, respectively, of the PCB **100** are illustrated showing the next manufacturing step. In this step, the drilling device **170** is advanced along the direction of arrow "B" so that the distal end **172** of the drilling device **170** contacts and engages the lower end **154** of the plating **150** in the VIA **140**.

[0044] Turning to FIGS. **9**A and **9**B, a top perspective view and a side view, respectively, of the PCB **100** are illustrated showing the next manufacturing step. In this step, the drilling device **170** has been advanced along the direction of arrow "B" a distance "**d1**" into the VIA **140**. As the drilling device **170** rotates along the direction of arrow "A" and moves along the direction of arrow "B", a portion of the plating **150** is removed from the VIA **140** due to the movement of the drilling device **170**, which is the backdrilling process.

[0045] Turning to FIGS. **10**A and **10**B, a top perspective view and a side view, respectively, of the PCB **100** resulting from the step illustrated in FIGS. **9**A and **9**B are shown. The backdrilling process results in the plating **150** extending from its upper end **152** to an intermediate end **155**, which is located at a position between the upper end **152** and the previously illustrated lower end **154** of the plating **150**. As shown, a portion of the plating **150** has been removed by the backdrilling process, which results in the VIA **140** having a backdrilled portion **149**A and a non-backdrilled portion **149**B. The backdrilled portion **149**A of the VIA **140** reveals the inner surface **146** of the VIA **140** which defines the channel **148** of the VIA **140**.

[0046] In this implementation, the intermediate end **155** of the plating **150** resulting from the backdrilling is located below the connection of high-speed trace **132** to the plating **150** of the VIA **140**. High-speed traces **130** and **132** are both connected to the plating **150**. A signal can travel along the direction of arrow "C" along high-speed trace **130**, along the direction of arrow "D" along plating **150**, and then along the direction of arrow "E" along high-speed trace **132**.

[0047] Turning to FIGS. **11**A and **11**B, a top perspective view and a side view, respectively, of PCB **100** showing the next manufacturing step are illustrated. In this manufacturing step, a supplying device **180** that contains a filling material **190** is positioned proximate to the lower end **144** of the VIA **140**. The supplying device **180** is actuated to insert the filling material **190** into the backdrilled portion **149**A, and in particular, into the channel **148** of the VIA **140**.

[0048] In FIGS. **12**A and **12**B, an intermediate filling step is illustrated. As shown, the filling material **190** continually inserted into the VIA **140**. The filling material **190** moves from the backdrilled portion **149**A into the non-backdrilled portion **149**B as more filling material is supplied.

[0049] Referring to FIGS. **13**A and **13**B, a top perspective view and a side view, respectively, of PCB **100** showing the next manufacturing step are illustrated. In this manufacturing step, the supplying device **180** continues to provide additional filling material **190** into the VIA **140**. As shown, a sufficient amount of filling material **190** is inserted into the VIA **140** so that the entire VIA **140** is filled with the filling material **190**. In particular, the filling material **190** fills up the backdrilled portion **149**A of the VIA **140** and the non-backdrilled portion **149**B of the VIA **140**.

[0050] As shown in FIG. **13**B, the filling material **190** has two portions in the VIA **140**. The filling material **190** has a lower portion **192** located in the backdrilled portion **149**A and an upper portion **194** located in the non-backdrilled portion **149**B.

[0051] In some aspects, the techniques described herein relate to a printed circuit board, comprising at least one layer; a VIA extending through the at least one layer, the VIA being backdrilled and filled with a backdrill filling material; and at least one trace connected to the VIA, wherein the backdrill filling material is selected based on backdrill filling material dielectric constant ("dk") properties to reduce impedance mismatches at a transition between the VIA and the at least one trace.

[0052] In one aspect, the backdrill filling material is selected based on its dk properties to improve insertion loss in the PCB.

[0053] In another aspect, the backdrill filling material is selected based on its dk properties to improve return loss in the PCB.

[0054] In yet another aspect, the backdrill filling material is selected based on its dk properties to reduce signal reflections between the VIA and the at least one trace.

[0055] In one aspect, the backdrill filling material has a dk less than 4.4.

[0056] In a further aspect, the backdrill filling material has a dk of 3.5 or less.

[0057] In another further aspect, the backdrill filling material has a dk of 2.5 or less.

[0058] In another aspect, the backdrill filling material is selected based on its dk properties to reduce signal reflections between the VIA and the at least one trace, and to improve return loss in the PCB.

[0059] In yet another aspect, the VIA is a high-speed VIA, and the at least one trace is a high-speed trace.

[0060] In some aspects, the techniques described herein relate to a printed circuit board, comprising at least one layer; a high-speed VIA extending through the at least one layer; and at least one trace connected to the high-speed VIA, wherein the high-speed VIA is backdrilled and filled with a backdrill filling material that is selected based on backdrill filling material dk properties to minimize any effect of mismatch of impedances at a transition between the high-speed VIA and the at least one trace.

[0061] In one aspect, the backdrill filling material is selected based on its dk properties to improve insertion loss in the printed circuit board.

[0062] In an alternative aspect, the backdrill filling material is selected based on its dk properties to improve return loss in the printed circuit board.

[0063] In another aspect, the backdrill filling material is selected based on its dk properties to reduce signal reflections between the high-speed VIA and the at least one trace.

[0064] In one aspect, the backdrill filling material has a dk less than 4.4.

[0065] In a further aspect, the backdrill filling material has a dk of 3.5 or less.

[0066] In another further aspect, the backdrill filling material has a dk of 2.5 or less.

[0067] In some aspects, the techniques described herein relate to a printed circuit board, comprising at least one layer; a high-speed VIA extending through the at least one layer; and at least one trace connected to the high-speed VIA, wherein the high-speed VIA is backdrilled and filled with a backdrill filling material that is selected based on backdrill filling material dk properties to minimize any mismatching of impedances between the high-speed VIA and the at least one trace.

[0068] In one aspect, the backdrill filling material is selected based on its dk properties to improve either insertion loss or return loss in the printed circuit board.

[0069] In another aspect, the backdrill filling material is selected based on its dk properties to reduce signal reflections between the high-speed VIA and the at least one trace. In yet another aspect, the backdrill filling material has a dk less than 4.4.

Variations and Implementations

[0070] Note that in this specification, references to various features (e.g., elements, structures, nodes, modules, components, engines, logic, steps, operations, functions, characteristics, etc.) included in "one embodiment," "example embodiment," "an embodiment," "another embodiment," "certain embodiments," "some embodiments," "various embodiments," "other embodiments," "alternative embodiment," and the like are intended to mean that any such features are included in one or more embodiments of the present disclosure, but may or may not necessarily be combined in the same embodiments.

[0071] It is also noted that the operations and steps described with reference to the preceding figures illustrate only some of the possible scenarios that may be executed by one or more entities discussed herein. Some of these operations may be deleted or removed where appropriate, or these steps may be modified or changed considerably without departing from the scope of the presented concepts. In addition, the timing and sequence of these operations may be altered considerably and still achieve the results taught in this disclosure. The preceding operational flows have been offered for purposes of example and discussion. Substantial flexibility is provided by the embodiments in that any suitable arrangements, chronologies, configurations, and timing mechanisms may be provided without departing from the teachings of the discussed concepts.

[0072] As used herein, unless expressly stated to the contrary, use of the phrase "at least one of," "one or more of," "and/or," variations thereof, or the like are open-ended expressions that are both conjunctive and disjunctive in operation for any and all possible combination of the associated listed items. For example, each of the expressions "at least one of X, Y and Z," "at least one of X, Y or Z," "one or more of X, Y and Z," "one or more of X, Y or Z" and "X, Y and/or Z" can mean any of the following: 1) X, but not Y and not Z; 2) Y, but not X and not Z; 3) Z, but not X and not Y; 4) X and Y, but not Z; 5) X and Z, but not Y; 6) Y and Z, but not X; or 7) X, Y, and Z.

[0073] Each example embodiment disclosed herein has been included to present one or more different features. However, all disclosed example embodiments are designed to work together as part of a single larger system or method. This disclosure explicitly envisions compound embodiments that combine multiple previously-discussed features in different example embodiments into a single system or method.

[0074] Additionally, unless expressly stated to the contrary, the terms "first," "second," "third," etc., are intended to distinguish the particular nouns they modify (e.g., element, condition, node, module, activity, operation, etc.). Unless expressly stated to the contrary, the use of these terms is not intended to indicate any type of order, rank, importance, temporal sequence, or hierarchy of the modified noun. For example, "first X" and "second X" are intended to designate two "X" elements that are not necessarily limited by any order, rank, importance, temporal sequence, or hierarchy of the two elements. Further as referred to herein, "at least one of" and "one or more of" can be represented using the "(s)" nomenclature (e.g., one or more element(s)).

[0075] The above description is intended by way of example only. Although the techniques are illustrated and described herein as embodied in one or more specific examples, it is nevertheless not intended to be limited to the details shown, since various modifications and structural changes may be made within the scope and range of equivalents of the claims.

What is claimed is:

1. A printed circuit board (PCB), comprising:

at least one layer;

a VIA extending through the at least one layer, the VIA being backdrilled and filled with a backdrill filling material; and

at least one trace connected to the VIA, wherein the backdrill filling material is selected based on backdrill filling material dielectric constant ("dk") properties to reduce impedance mismatches at a transition between the VIA and the at least one trace.

2. The printed circuit board of claim **1**, wherein the backdrill filling material is selected based on its dk properties to improve insertion loss in the PCB.

3. The printed circuit board of claim **1**, wherein the backdrill filling material is selected based on its dk properties to improve return loss in the PCB.

4. The printed circuit board of claim **1**, wherein the backdrill filling material is selected based on its dk properties to reduce signal reflections between the VIA and the at least one trace.

5. The printed circuit board of claim **1**, wherein the backdrill filling material has a dk less than 4.4.

6. The printed circuit board of claim **5**, wherein the backdrill filling material has a dk of 3.5 or less.

7. The printed circuit board of claim **6**, wherein the backdrill filling material has a dk of 2.5 or less.

8. The printed circuit board of claim **1**, wherein the backdrill filling material is selected based on its dk properties to reduce signal reflections between the VIA and the at least one trace, and to improve return loss in the PCB.

9. The printed circuit board of claim **1**, wherein the VIA is a high-speed VIA, and the at least one trace is a high-speed trace.

10. A printed circuit board, comprising:

at least one layer;

a high-speed VIA extending through the at least one layer; and

at least one trace connected to the high-speed VIA, wherein the high-speed VIA is backdrilled and filled with a backdrill filling material that is selected based on backdrill filling material dk properties to minimize any effect of mismatch of impedances at a transition between the high-speed VIA and the at least one trace.

11. The printed circuit board of claim **10**, wherein the backdrill filling material is selected based on its dk properties to improve insertion loss in the printed circuit board.

12. The printed circuit board of claim **10**, wherein the backdrill filling material is selected based on its dk properties to improve return loss in the printed circuit board.

13. The printed circuit board of claim **10**, wherein the backdrill filling material is selected based on its dk properties to reduce signal reflections between the high-speed VIA and the at least one trace.

14. The printed circuit board of claim **10**, wherein the backdrill filling material has a dk less than 4.4.

15. The printed circuit board of claim **14**, wherein the backdrill filling material has a dk of 3.5 or less.

16. The printed circuit board of claim **15**, wherein the backdrill filling material has a dk of 2.5 or less.

17. A printed circuit board, comprising:

at least one layer;

a high-speed VIA extending through the at least one layer; and

at least one trace connected to the high-speed VIA, wherein the high-speed VIA is backdrilled and filled with a backdrill filling material that is selected based on backdrill filling material dk properties to minimize any mismatching of impedances between the high-speed VIA and the at least one trace.

18. The printed circuit board of claim **17**, wherein the backdrill filling material is selected based on its dk properties to improve either insertion loss or return loss in the printed circuit board.

19. The printed circuit board of claim **17**, wherein the backdrill filling material is selected based on its dk properties to reduce signal reflections between the high-speed VIA and the at least one trace.

20. The printed circuit board of claim **17**, wherein the backdrill filling material has a dk less than 4.4.

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